

Ayushman Kumar

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Professional Summary

- B.Tech AI & Data Science student at IIT Patna seeking backend engineering and systems design roles. Strong foundation in full-stack development, API design, and cryptographic security. Eager about building scalable systems and solving complex technical problems through continuous learning and hands-on development.

Education

- IIT Patna – B.Tech Artificial Intelligence & Data Science Aug 2024 – Jul 2028
CGPA: 8.96/10
- M.D.J Public School – Higher Secondary (PCM): 96%, Class X: 97.5%

Projects

- Institutional CMS – IIT Patna React, Node.js, Prisma, PostgreSQL, Docker
 - Engineered platform serving **100+ faculty** across **10+ departments**; built granular RBAC achieving **100% data consistency**
 - Containerized via Docker and deployed on institutional infrastructure with Nginx; optimized for high availability and infrastructure resilience
 - Reduced administrative overhead by **60%** through enabled permission-based faculty self-service editing
 - Clinical Data Normalization Engine – JILO Health (2nd Place) Python, LangChain, React, PostgreSQL, pgvector
 - Architected Revenue Cycle Management pipeline processing unstructured patient reports; designed normalization to FHIR-compliant JSON with ICD-10 and CPT mapping
 - Achieved **87% semantic accuracy** on clinical entity recognition; reduced preauthorization turnaround from **2+ hours to 8 minutes** per case
 - Built retrieval-augmented generation pipeline with semantic search; streamlined insurance preauthorization form population across **50+ claim fields**
 - SecureVote – Decentralized Voting Platform MERN, OAuth 2.0, Ring Signatures
 - Engineered cryptographic voting protocol using Linkable Ring Signatures; guaranteed **100% voter anonymity** and prevented duplicate ballots
 - Deployed secure OAuth 2.0 authentication with custom middleware and RBAC; managed election state transitions
 - Achieved **sub-100ms** cryptographic verification per ballot with integrity validation for all records
- Demo | GitHub

Skills

- Languages: Python, JavaScript (ES6+), C/C++, SQL, Go
- Backend & Data: Node.js, Express.js, FastAPI, RESTful API Design, PostgreSQL, MongoDB, Prisma ORM, pgvector
- Frontend & Infrastructure: React, Next.js, Docker, Kubernetes, Nginx, CI/CD, System Architecture
- AI & ML: LangChain, PyTorch, Semantic Search, Vector Embeddings, Prompt Engineering, RAG Systems, NLP
- Security & Compliance: Cryptography, OAuth 2.0, Role-Based Access Control (RBAC), Secure Authentication, Data Privacy

Leadership & Extracurricular

- Web Development Lead – Google Developer Groups (GDG), IIT Patna: Co-organized **8+ technical events** including study jams and hackathons; directed workshops on full-stack architecture and API design; mentored **20+ peers** in production deployment patterns
- Core Team Member – NJack (Technical Club, IIT Patna): Conducted **8+ workshops** for freshers covering competitive programming and software engineering fundamentals; contributed to technical event execution
- Flagship Coordinator – Celesta: Managed logistics for **major institute tech festival**; coordinated teams overseeing **100+ participants**